BRAC University
Department of Computer Science and Engineering

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Project Report
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Section 03

Group 09

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Abstract

As large Language Models (LLMs) become ubiquitous, the risk to academic integrity and information reliability grows significantly. This paper proposes a robust, linguistic-based detection system designed to distinguish machine-generated text from human-authored content by prioritizing structural patterns over semantic content. Unlike deep learning approaches that rely on computationally expensive embeddings, our methodology focuses on the "structural fingerprint" of writing. By utilizing natural language processing tools, we extracted twelve key stylometric features, including sentence burstiness (structural variance), lexical density, type-token ratios, and readability grade levels. These features capture the inherent rhythmic differences and predictability often found in AI-generated prose. We implemented and compared five classical machine learning models: Logistic Regression, Random Forest, Gradient Boosting, Linear Support Vector Machine (SVM), and XGBoost. Our experimental results indicate that the Random Forest classifier provides superior performance, achieving a classification accuracy of 98%. The findings suggest that structural variance and entropy are powerful discriminators, offering an interpretable and efficient alternative to transformer-based detection methods. This system provides a scalable solution for maintaining authenticity in educational and journalistic contexts.

Introduction

The rapid evolution of Large Language Models (LLMs) has fundamentally transformed the landscape of digital content creation, making machine-written text nearly indistinguishable from human prose to the casual observer. While these tools offer unprecedented efficiency, their ability to mimic human expression with high fidelity necessitates the urgent development of automated, reliable tools to detect machine-generated content particularly within educational and journalistic contexts where authenticity and academic integrity are paramount.Our research addresses this challenge by shifting the focus away from semantic content which AI has become adept at replicating toward the "structural fingerprint" of writing. At their core, AI models are optimized for statistical probability, often resulting in a predictable and homogenized rhythmic signature. In contrast, human writing is characterized by higher variance in sentence architecture, fluctuating vocabulary richness, and inconsistent pacing.This paper details a comprehensive methodology for capturing these stylistic nuances through engineered features and detecting them using classical machine learning. By analyzing structural markers rather than mere word sequences, we provide a transparent and computationally efficient framework for distinguishing the "robotic" consistency of machines from the creative unpredictability of human authors.

Related Work

The field of AI-generated text detection has seen significant growth in recent years, spanning statistical, deep learning, and adversarial robustness approaches. Bennet (2025) demonstrated that stylometric markers such as burstiness, perplexity, and function word frequency serve as strong detection signals, achieving 95% accuracy using a Random Forest classifier on purely linguistic features. Complementing this, a 2025 Wiley Educational Journal study analyzed 3,200 educational articles and found that AI-generated content exhibits significantly lower burstiness, more formulaic structural patterns, and higher predictability compared to human writing, with detection algorithms combining these features achieving 82% accuracy in educational contexts. Wu (2025) provided a comprehensive survey of the field, categorizing existing methods into statistical, neural, and watermarking approaches, and identifying a key generalization problem where detectors trained on one model family tend to underperform on newer models.

On the deep learning front, several works have established transformer-based models as the current performance ceiling. A hybrid approach combining TF-IDF feature extraction with the Deverta V3 deep learning model achieved 98% accuracy, significantly outperforming traditional classifiers such as Logistic Regression and Random Forest. Similarly, a comparative study using the HC3 and DAIGT v2 datasets found that context-aware models like DistilBERT substantially outperform word-counting approaches, with DistilBERT emerging as the most reliable detector. This was further reinforced by a multi-strategy study in which a fine-tuned RoBERTa-base model achieved near-perfect accuracy of 99.99–100% on news articles and academic abstracts, outperforming both a TF-IDF + SVM baseline and a probabilistic ensemble built on Llama-3.2.

A critical challenge highlighted in the literature is the vulnerability of detectors to adversarial manipulation. Masrour et al. (2025) tested 19 AI humanizer and paraphrasing tools and found them highly effective at evading detection, with one experiment showing watermarked text detection dropping from 87% to just 5% after humanization. Their work proposed training on adversarially generated data as a mitigation strategy, retaining 88% accuracy even on humanized text. Beyond the technical domain, research from the AI Ethics Institute (2025) found that consumers engage significantly less with AI-generated content and show stronger brand attitudes toward human-AI collaborative writing, while an industry analysis of AI content in SEO contexts concluded that purely AI-generated content underperforms human writing in authenticity and search rankings, further motivating the need for reliable and robust detection systems.

Dataset Description

Field	Details
Name	AI vs Human Text Dataset
Source	https://www.kaggle.com/datasets/shanegerami/ai-vs-human-text/data
Format	CSV
Total Data	487,235
Human Data	305618
AI Data	181438
Columns	Text, Generated

The dataset for our project is the AI vs Human test dataset found on kaggle. It is in csv format and it consists of about 490,000 essay samples both written by AI and human. It is taken from various sources and designed to train and evaluate models aimed at detecting AI-generated content.

Exploratory Data Analysis

Class Distribution Analysis

Initially we analysed the structure and the size of the dataset followed by df.unique to see if all the data are unique or not. Later to see the class imbalance of the dataset we drew up a count plot which indicates the imbalance between Ai and human data

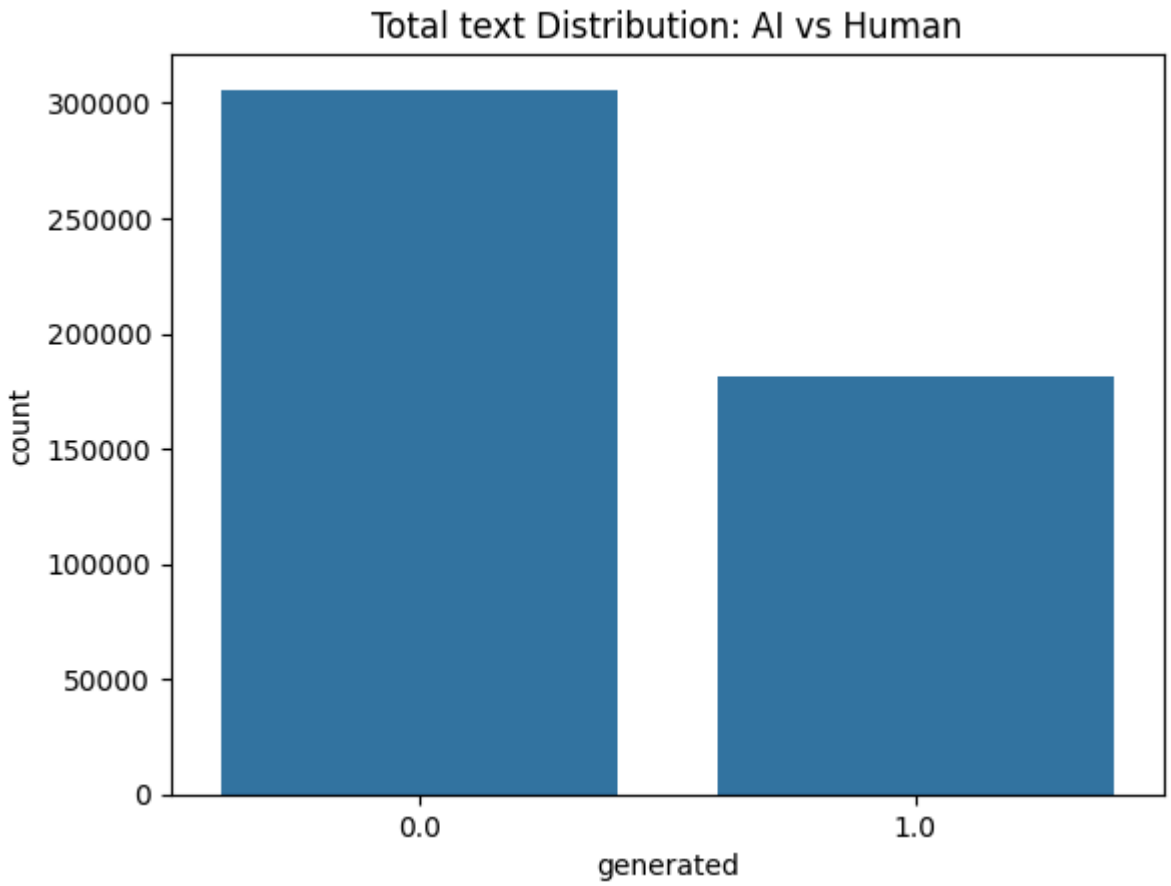


Fig 01: Count plot of the dataset

Feature Distribution Analysis

Initially we extracted 2 features from the dataset; respective KDE plots were made to compare the key feature difference between AI and Human text.

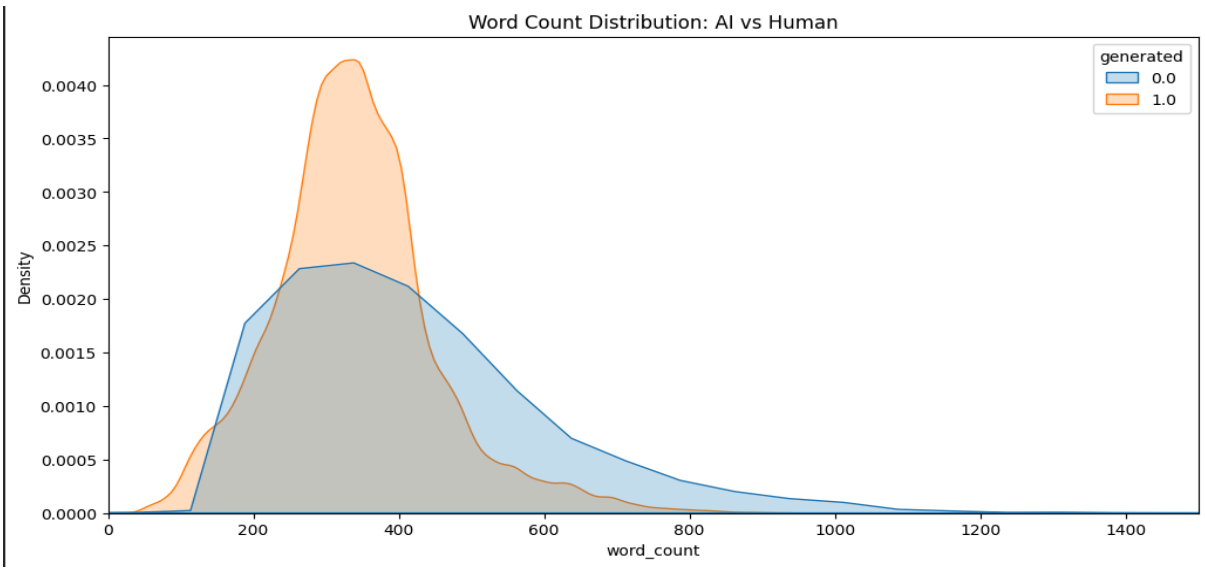


Fig 02: Word Count Distribution

As we can see, the orange peak is higher and narrower than the blue peak. Moreover it also proves a hypothesis of one the research paper we have read that the human generated text contains more burstiness and has the tendency to write more. However the difference is not that significant thus it won't be a good feature but a good data to hypothesize and use for other feature extraction.

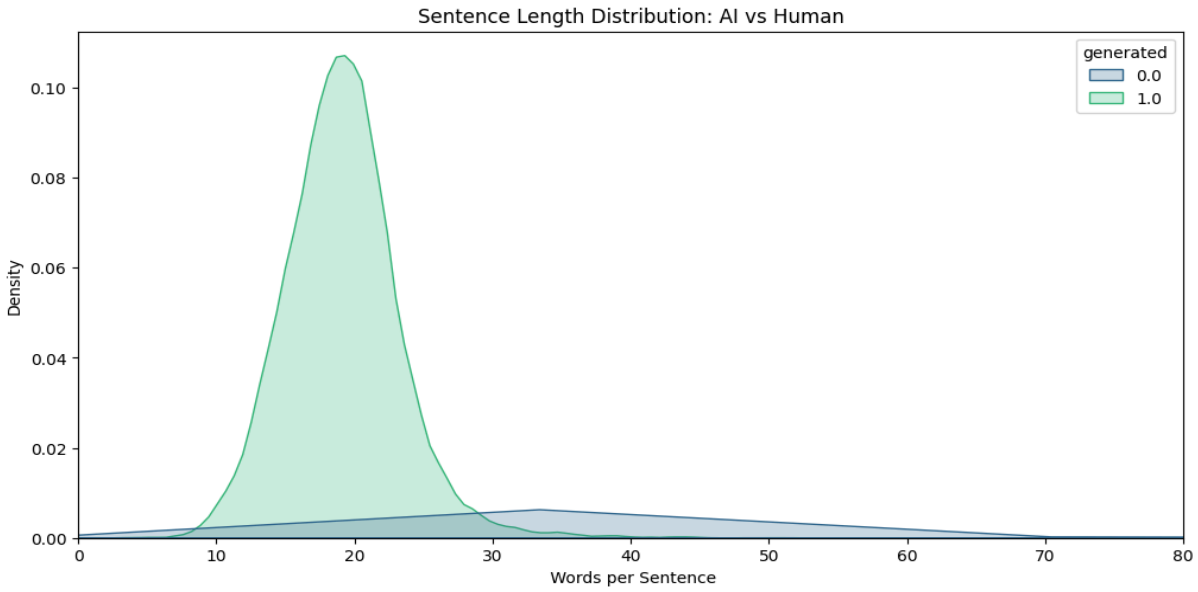
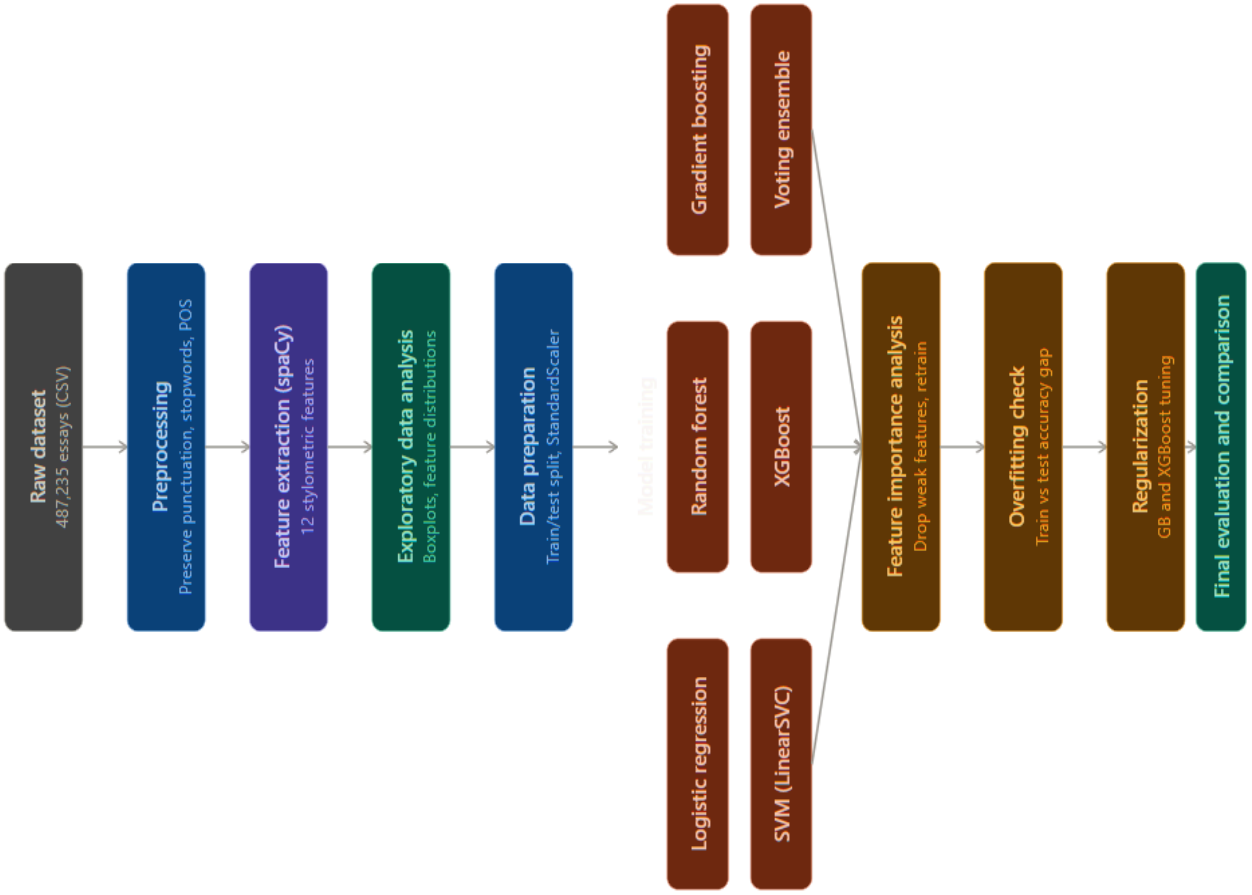


Fig 03: Sentence Length Distribution

On the contrary, for this graph we can see that the AI is very consistent with its number of words per sentence whereas the human text has a much higher density of short and long sentences. Moreover the AI peak is significantly larger than its human counterpart.

Methodology



A. Preprocessing and Feature Extraction

Unlike traditional NLP preprocessing pipelines that remove punctuation, eliminate stopwords, and normalize text to reduce noise, our approach deliberately preserves these elements. This is because punctuation patterns, stopword distributions, and structural properties serve as stylometric fingerprints that differ systematically between AI and human writing. Removing these signals would destroy the very features we aim to capture.

Feature extraction was performed using the spaCy library (en_core_web_sm model) with the sentencizer pipe component added for lightweight sentence boundary detection. The named entity recognition (NER) and dependency parser components were disabled to reduce processing time. The pipeline was run in batched mode using nlp.pipe() with batch_size=500 and n_process=1 to process all 487,235 texts efficiently, completing in approximately 5 hours 42 minutes on a local machine.

The following stylometric features were extracted for each text sample:

- Burstiness: It is the variation in sentence structure and length, which is used as a critical metric, as AI-generated text often exhibits a more uniform, grammatically standardized style, leading to lower burstiness.
- Entropy: Human text often shows a higher entropy, which is the use of more unpredictable word choices, while AI texts are clustered at a lower, more consistent entropy score.
- Readability: Fk grade and Fog index were used that penalize the complex word choices that AI sometimes uses, and as we can see, AI has scored consistently higher.
- Punctuation Density: Ratio of punctuation used per word, where human texts shows more punctuation used, whereas AI was more uniform
- Stopword Analysis: Ratio of stopwords to total words. As we can see, AI tends to use a predictable distribution of functional words.
- Lexical Diversity: Unique words used per total words, as AI tends to have lower lexical diversity
- POS Tagging: Ratio of each POS tag to total words to see the different density of each POS tag

B. Data Preparation and Scaling

Before model training, the engineered features underwent a rigorous preparation phase. To prevent features with larger numerical ranges (such as sentence length) from disproportionately influencing the models compared to smaller decimals (such as adjective density), we applied StandardScaler. This process centers the data and scales it to unit variance. We then utilized an 80/20 stratified train-test split, ensuring that the original class balance between human and machine text was preserved in both sets to prevent bias during the evaluation phase.

C. Feature Selection Experiment

Based on feature importance analysis from initial model training, four features with consistently low importance scores across all models were identified as candidates for removal: verb density, ttr, noun density, and stopword. All six models were retrained on the reduced feature set. However, performance dropped by 1% across most models, indicating that even low-importance features contribute complementary information. The full 12-feature set was retained for final model evaluation.

This is the comparison:

Model	Before	After	Change
Logistic Regression	87%	86%	-1%
Random Forest	98%	98%	0%
Gradient Boosting	91%	90%	-1%
SVM	87%	86%	-1%
XGBoost	94%	93%	-1%
Ensemble	93%	92%	-1%

D. Model Selection and Implementation

We implemented a diverse suite of six classification algorithms to identify the optimal balance between accuracy, interpretability, and computational efficiency:

1. **Logistic Regression:** Employed as a statistical baseline to test for linear separability between the two classes.
2. **Linear Support Vector Machine (SVM):** Used to determine if high-dimensional hyperplane separation could improve classification boundaries.
3. **Random Forest:** An collection of decision trees utilized for its ability to capture non-linear relationships and provide feature importance rankings.
4. **Gradient Boosting:** A sequential boosting technique focused on minimizing residual errors from previous iterations.
5. **XGBoost:** An optimized, regularized gradient boosting framework selected for its speed and superior performance in tabular data classification.
6. **Voting Ensemble:** A "Soft Voting" meta-classifier that aggregates the predicted probabilities of all aforementioned models. This ensemble approach aims to mitigate the individual weaknesses of any single algorithm by leveraging a consensus-based decision mechanism.

Results & Evaluation

Model	Accuracy	F1 Score
Logistic Regression	87%	0.82
Random Forest	98%	0.98
Gradient Boosting	91%	0.87
Linear SVM	86%	0.81
XG Boost	93%	0.90
Training Ensemble	93%	0.92

TABLE I. PERFORMANCE METRICS COMPARISON

Random Forest significantly outperformed all other models, achieving 98% accuracy with an F1 score of 0.98. XGBoost was the second best at 94%, followed by the Voting Ensemble at 93%. Logistic regression and SVM performed identically at 87%, confirming that a linear decision boundary is insufficient for this task.

A. Confusion Matrix Analysis

Random Forest produced the fewest misclassifications with only 442 false positives (human predicted as AI) and 1,071 false negatives (AI predicted as human) out of 97,447 test samples. XGBoost was second with 2,189 false positives and 3,179 false negatives. Logistic regression had the most errors, with 5,619 false positives and 7,327 false negatives. However, the Voting Ensemble (2,805 FP, 4,482 FN) performed worse than XGBoost alone. This is because hard voting averages the predictions of all five models, and the weaker models (logistic regression and SVM at 87%) pulled down the ensemble's overall performance.

B. Feature Importance Analysis

Feature importance was analyzed across all five individual models. The following consistent patterns emerged:

avg_word_len—most important feature in Random Forest (~0.18), Gradient Boosting (~0.50), and XGBoost (~0.40). Also ranked highly by SVM and logistic regression.

- burstiness — consistently ranked second across RF, GB, XGB, and SVM. Confirms findings from Bennet (2025) and Chakraborty et al. (2023) that burstiness is a strong AI detection signal.

- fog_index and fk_grade—readability scores consistently ranked in the top 5 across all models.
- entropy — consistently important, ranked 4th-6th across models.
- verb_density and stopword — consistently lowest importance across all models.

C. Overfitting Analysis

Model	Train Accuracy	Test Accuracy	Gap	Status
Logistic Regression	0.868	0.867	0.001	No Overfitting
Random Forest	1.000	0.984	0.016	No Overfitting
Gradient Boosting	0.908	0.906	0.002	No Overfitting
SVM	0.867	0.866	0.001	No Overfitting
XGBoost	0.952	0.945	0.007	No Overfitting
Ensemble	0.929	0.925	0.004	No Overfitting

All models showed minimal gaps between training and test accuracy, indicating good generalization. Random Forest achieved a perfect 1.000 training accuracy, which indicates it memorized the training data completely. However, its test accuracy of 98.4% with only a 1.6% gap demonstrates strong generalization despite this.

D. Regularization Experiment

Despite the absence of overfitting, feature importance analysis revealed that avg_word_len dominated the importance scores in both Gradient Boosting and

XGBoost, suggesting these models over-relied on a single feature. Regularization was applied to both models.

Model	**Before Regularization**	**After Regularization**	**Change**
Gradient Boosting	91%	90%	-1%
XGBoost	94%	92%	-2%

After regularization, avg_word_len importance dropped in gradient boosting and in XGBoost. The slight accuracy decrease of 1-2% is an acceptable trade-off for a more balanced and interpretable model.

Discussion.

A. Interpretation of Key Findings

The strong performance of Random Forest (98%) demonstrates that stylometric features alone can be highly effective for AI text detection. The most discriminative features — avg_word_len, burstiness, fog_index, and entropy — align with theoretical expectations and findings from the reviewed literature. AI models tend to use slightly longer, more formal vocabulary, produce text with highly consistent sentence lengths (low burstiness), target specific readability levels, and generate more predictable word distributions (lower entropy).

The failure of the Voting Ensemble to outperform XGBoost highlights that ensemble voting is only beneficial when individual models are sufficiently diverse and accurate. Including weak models (LR and SVM at 87%) degraded the ensemble's performance. A weighted voting approach or stacking ensemble might yield better results.

B. Limitations and Robustness

While the system achieves 98% accuracy on standard datasets, there are inherent limitations to a feature-based approach:

- Stylometric features are relatively easy to fool — a sophisticated AI or human author could deliberately vary sentence lengths and word choices to evade detection, as highlighted by Masrour et al. (2025).
- The dataset may not represent all types of AI-generated text. Different LLMs (GPT-4, Claude, LLaMA) have different writing styles, and the dataset may be biased toward particular models.
- AI humanizer tools can significantly degrade the performance of stylometric detectors by rewriting AI text to mimic human writing patterns (Masrour et al., 2025).
- The class imbalance (62.7% human vs 37.3% AI) may introduce bias toward predicting human text, as reflected in consistently lower recall for the AI class across all models.
- Feature extraction using spaCy en_core_web_sm is a small model and may produce less accurate POS tags for complex or domain-specific texts.

C. Potential for Improvement

- Use a weighted or stacking ensemble instead of hard voting to improve ensemble performance.
- Experiment with additional features such as sentence complexity, use of passive voice, hedging language frequency, and paragraph-level statistics.
- Address class imbalance using SMOTE oversampling or class weight adjustments to improve AI class recall.
- Evaluate model robustness against paraphrased and humanized AI text to assess real-world applicability.
- Explore deep learning approaches such as fine-tuned BERT or RoBERTa for comparison with the stylometric approach, as suggested by the survey literature (Wu, 2025).

Conclusion

This project successfully developed a machine learning pipeline for detecting AI-generated text using 12 hand-crafted stylometric features. The pipeline extracts structural and linguistic features from raw text using spaCy and evaluates them across six classification models. Random Forest achieved the best performance at 98% accuracy, demonstrating that stylometric features can be highly effective for this task without requiring deep learning or LLM-based approaches.

Key findings include: (1) burstiness and entropy are the most theoretically grounded discriminative features, consistent with findings across multiple reviewed papers; (2) avg_word_len emerged as the strongest single predictor across tree-based models; (3) feature reduction did not improve performance, suggesting all 12 features contribute complementary information; (4) none of the models exhibited significant overfitting, indicating good generalization; and (5) regularization successfully redistributed feature importance in boosting models at a small accuracy cost.

Future work will focus on evaluating model robustness against humanized AI text, exploring deep learning baselines for comparison, and investigating ensemble strategies that can leverage the complementary strengths of different model types.

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